

Hyuna Jeon

M.S. Student @ Hanyang University

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Research Interests

Large Language Models for time-series reasoning, multimodal recommendation, ontology- and knowledge-graph-based RAG, AI agents, and data-centric AI. My research focuses on structuring heterogeneous context and enabling LLMs to retrieve, select, and reliably use the information most relevant to a given query or situation.

Education

Hanyang University, Seoul, Republic of Korea

Mar. 2025 – Present

M.S. in Artificial Intelligence

– IDSL Lab, Advisor: Prof. Misuk Kim

– Expected graduation: Feb. 2027; GPA: 4.06 / 4.5

Kyonggi University, Suwon, Republic of Korea

Mar. 2018 – Feb. 2023

B.S. in Applied Statistics

– GPA: 3.83 / 4.5

Publications

[C.1] Bridging Time Series and Language via Query-Adaptive Multi-Granularity Routing for Temporal Reasoning

Hyuna Jeon*, Jaeyoung Lee*, and Misuk Kim

EMNLP 2026 (Conference on Empirical Methods in Natural Language Processing), Main Conference (Acceptance Rate: 15.4%)

[C.2] Ontology-Based Knowledge Graphs for Industrial Standards with Scope-Aware Logical Rule Structuring

Jiin Park*, Hyuna Jeon*, Yoonseo Lee, and Misuk Kim

EMNLP 2026 (Conference on Empirical Methods in Natural Language Processing), Main Conference (Acceptance Rate: 15.4%)

[J.1] BearGen: LLM-Guided Signal Generation Framework for Bearing Fault Diagnosis

Jaeyoung Lee, Hyuna Jeon, Uiin Kim, and Misuk Kim

Advanced Engineering Informatics, 2026 (SCIE; Impact Factor: 11.5; JCR Top 2%)

* Equal contribution.

Research and Professional Experience

• IDSL Research Lab, Hanyang University

Nov. 2024 – Present

Graduate Researcher

Seoul, Republic of Korea

- Conduct research on time-series LLMs, multimodal recommendation, and ontology/KG-RAG, with a focus on enabling LLMs to understand and use heterogeneous context.
- Developed query-adaptive multi-granularity routing that selects Local, Global, and Cross temporal information according to the question and aligns the selected representations with an LLM.
- Built ontology-based knowledge graphs for industrial standards that preserve document hierarchy, rule scope, condition-consequence logic, and evidence distributed across text, tables, and footnotes.
- Designed expert-based multimodal recommendation methods that estimate modality quality and remain robust when image, text, or audio information is noisy or missing.
- Research outputs include two papers accepted at EMNLP 2026, one journal article in *Advanced Engineering Informatics*, and two related patent applications.

• The IMC

Apr. 2023 – Aug. 2024

Researcher, AI Research & Modeling Team

Seoul, Republic of Korea

- Developed smart-farm environmental forecasting models using Transformer and ARIMA, and device-operation prediction models using Random Forest.
- Researched and applied Kriging, Kalman Filter, and AutoML for greenhouse optimization; monitored control outcomes and maintained deployed models.
- Contributed to an LLaMA3-based chatbot for the Gyeonggi Economic Platform and improved domain question answering through supervised fine-tuning.
- Improved Python web-crawling pipelines and managed SQL/NoSQL data ingestion and monitoring for a marketing intelligence platform.
- Supported AI/data proposals for defense and local-government projects and assisted with AI/big-data training for military personnel.

• Ministry of the Interior and Safety / NIA

Sep. 2020 – Dec. 2020

Public Data Intern

Republic of Korea

- Performed public-data quality diagnosis, data cleaning, online verification, and support for newly released public datasets.

Selected Research Projects

- **TiMER: Query-Adaptive Multi-Granularity Routing for Temporal Reasoning**

Python, PyTorch, LLM, Time-Series, Query-Adaptive Routing

- Designed a multi-granularity time-series encoder and routing framework that captures fine-grained local patterns, coarse-grained global trends, and cross-granularity dependencies.
 - Developed query-guided routing to select the temporal information most relevant to each question and align it with LLM representations for time-series question answering.
 - Evaluated both predictive performance and model efficiency through ablation studies and comparisons of parameter and training/inference costs.

2026
- **Ontology-Based KG-RAG for Industrial Standards**

Python, LLM, RAG, Knowledge Graph, Ontology, Neo4j

- Extended knowledge triples to preserve Section-Subsection hierarchy, rule scope, and condition-consequence relations in complex industrial standards.
 - Linked evidence distributed across text, tables, and footnotes to original document locations and enabled graph-based retrieval of applicable conditions and rules.
 - Built a scope-aware KG-RAG pipeline that retrieves grounded evidence for downstream question answering and reasoning.

2026
- **StateMoE: Robust Multimodal Recommendation under Mixed Modality Corruptions**

Python, PyTorch, Multimodal Learning, Recommendation, Mixture-of-Experts

- Designed a quality-adaptive multimodal recommendation framework that estimates the state of image, text, and audio modalities and remains robust under noisy or missing information.
 - Evaluated robustness across multiple corruption settings and studied expert-based correction mechanisms for degraded modalities.
 - Related work received a Best Paper Award from the Korean Data Mining Society in Aug. 2026.

2026

Additional Projects and Educational Content

- **SSAFY AI Educational Content Development | Upstage**

Data preprocessing and cleaning, data augmentation, data quality evaluation

- Developed lecture slides, hands-on code, and quizzes for graduates of the SSAFY software-engineering bootcamp.

Nov. 2024 – Dec. 2024
- **Hyundai Glovis Document Recognition Automation Proposal**

Object Detection, OCR, Ranking

- Conducted toy experiments on a multi-metric ranking method for object-detection and OCR outputs and contributed to the technical proposal.

Nov. 2024 – Dec. 2024

Technical Skills

- **Programming & Data:** Python, SQL, R, data preprocessing and analysis, web crawling
- **AI/ML:** PyTorch, TensorFlow, Keras, Scikit-learn, machine learning, deep learning, time-series modeling
- **LLM & Knowledge-Based AI:** LLM, SFT, RAG, Knowledge Graph, Ontology, Neo4j, AI agents
- **Recommendation & Multimodal AI:** Recommendation systems, multimodal learning, Mixture-of-Experts
- **Development & Databases:** Linux, Docker, SQL/NoSQL, Git/GitHub
- **Analytics & Visualization:** Statistical analysis, EDA, Tableau

Honors and Awards

- **Best Paper Award**

Korean Data Mining Society

- Awarded for research on quality-adaptive multimodal recommendation robust to noisy and missing modalities.

Aug. 25, 2026
- **Best Paper Award**

Korean Data Mining Society

- Awarded for an adaptive Mixture-of-Experts alignment framework that integrates graph embeddings for LLM-based recommendation.

Aug. 29, 2025

Training

- **Code States AI Bootcamp**

AI and Data Analysis Training, 1,120 hours

- Studied Python-based data analysis and preprocessing, machine learning, deep learning, NLP, SQL, and basic Flask/Docker workflows.
 - Conducted a Seoul single-person-household safety analysis and rent-prediction project, and separately developed a product recommendation system using user-item logs.

Aug. 2022 – Feb. 2023
Republic of Korea

Certifications and Language

- **Big Data Analysis Engineer**

Jul. 2024

• **SQL Developer (SQLD)**

Dec. 2022

• **Advanced Data Analytics Semi-Professional (ADsP)**

Mar. 2022

• **Korean History Proficiency Test, Level 1**

Aug. 2021

• **Survey Analyst, Level 2**

Jun. 2021

• **Computer Specialist in Spreadsheet & Database, Level 1**

Feb. 2021

• **TOEIC Speaking IM2 (120/200)**

Mar. 22, 2026

Volunteer Experience

- **LH Mentor and Child Mentoring Program**

Sep. 2019 – Feb. 2021

Academic mentoring and experiential learning support for an elementary-school student, 77 hours

- Co-mentored one elementary-school student with another mentor, providing academic guidance and experiential activities tailored to the student's learning level and interests.

Additional Information

GitHub: github.com/hyuna0926 **Technical Blog:** velog.io/@hyun26a__a